Machine Test Scoring Model – Zecpath

Objective

Provide a structured scoring model for evaluating practical software engineering skills through machine tests.

Evaluation Metrics

Metric	Weight
Correctness	40%
Runtime Efficiency	20%
Code Quality	20%
Problem Solving	20%

Final Score Formula

Task Score

```
Task Score =
(Correctness × 0.40) +
(Efficiency × 0.20) +
(Code Quality × 0.20) +
(Problem Solving × 0.20)
```

Overall Machine Test Score

```
Final Score =
(Task Score × 0.80) +
(Time Score × 0.20)
```

Time-Based Scoring

Time Used	Score
≤ 50% of limit	100

Time Used	Score
50–100%	70
Above limit	40

Performance Levels

Score	Result
85–100	Excellent
70–84	Good Performance
50–69	Average
Below 50	Needs Improvement

Example Output


```
{
  "candidate_id": "C5001",
  "task_score": 78.5,
  "time_score": 70,
  "final_score": 76.8,
  "decision": "Good Performance"
}
```


Benefits

- Objective evaluation
- Automated scoring
- Practical skill assessment
- Recruiter-friendly reports

Limitations

- Rule-based code quality analysis
- No plagiarism detection
- Limited optimization analysis

Future Improvements

- AI-powered code review
- AST-based analysis
- Code similarity detection
- Live coding evaluation